

Vibe Audio Protocol (V.A.P.) Logic Architecture v1.0

Scoring Systems & Rating Definitions

Phase I: Physical Analysis (DSP Layer)

Automated extraction of raw audio data using Digital Signal Processing.

Section 1: Structural Pillar (The Skeleton)

Subset 1.1: Percussive DNA

- **Metric: Kick Transient Profile (ms)**
 - *Logic:* Measurement of time (ms) from signal onset (-60dB) to Peak Amplitude (0dB).
 - *Scoring:*
 - **Sharp (Click):** < 10ms (Tech/Metal)
 - **Punch (Thud):** 10-30ms (Pop/Rock)
 - **Boom (Sub):** > 30ms (Trap/808)
- **Metric: Syncopation Index (0.0 - 1.0)**
 - *Logic:* Ratio of transient energy detected on weak beats vs. strong beats (1, 2, 3, 4).
 - *Scoring:* (OffBeat_Transients / Total_Transients)
 - 0.0 = Pure March (4/4 stomp)
 - 1.0 = Polyrhythmic Chaos (Jazz/Math Rock)

Section 2: Tonal Pillar (The Flesh)

Subset 2.1: Harmonic Profile

- **Metric: Dissonance Density (%)**
 - *Logic:* Percentage of total duration containing "high tension" intervals (Minor 2nd, Tritone, Major 7th).
 - *Scoring:*
 - **Consonant:** < 10% (Pop/Folk)
 - **Tense:** 10-40% (Jazz/Noir)
 - **Dissonant:** > 40% (Horrorcore/Avant-Garde)

Section 3: Timbral Pillar (The Skin)

Subset 3.1: Spectral Physics

- **Metric: Spectral Centroid (Hz)**

- *Logic*: The "center of gravity" of the frequency spectrum.
- *Scoring*:
 - **Dark/Muddy**: < 200Hz
 - **Warm/Body**: 200Hz - 2000Hz
 - **Bright/Airy**: > 2000Hz
- **Metric: Saturation Index (THD)**
 - *Logic*: Total Harmonic Distortion measurement relative to the fundamental frequency.
 - *Scoring*: 0.0 (Pure Sine) to 1.0 (White Noise/Bitcrush).

Phase II: Psychological Analysis (NLP & ML Layer)

Machine Learning models analyzing context, semantics, and human perception.

Section 4: Linguistic Pillar (The Voice)

Subset 4.1: Semantic Content

- **Metric: Explicit Severity Tier (1-4)**
 - *Logic*: Keyword density analysis against "Safety Lists" (Violence, Substance, Sexual).
 - *Scoring*:
 1. **Clean**: 0 matches.
 2. **Mild**: < 3 matches (Context dependent).
 3. **Explicit**: > 3 matches or specific "Hard" keywords.
 4. **Severe**: Hate speech or extreme violence markers (Auto-Filter).

Section 5: Affective Pillar (The Heart)

Subset 5.1: The Thayer Coordinates

- **Metric: Valence Score (-1.0 to +1.0)**
 - *Logic*: Sentiment analysis of lyrics + Major/Minor key determination.
 - *Scoring*:
 - -1.0 = Despair/Terror (Minor Key + Negative Sentiment)
 - 0.0 = Neutral/Ambient
 - +1.0 = Euphoria/Joy (Major Key + Positive Sentiment)
- **Metric: Arousal Score (0.0 to 1.0)**
 - *Logic*: RMS Amplitude (Loudness) + BPM + Spectral Density.
 - *Scoring*: 0.0 (Sleep) to 1.0 (Panic/Rage).

Section 6: Contextual Pillar (The Scene)

Subset 6.1: Scenario Probability

- **Metric: Scenario Confidence Score (%)**
 - *Logic*: Bayesian probability matching audio features to pre-defined clusters (e.g.,

- "Night Drive").
 - *Scoring*: $P(\text{NightDrive} \mid \text{Synthwave} + 100\text{bpm} + \text{MinorKey})$
 - 85% = Auto-Tag applied.

Phase III: Environmental & Social Analysis (I/O Layer)

Output formatting for hardware and community feedback loops.

Section 7: Photometric Pillar (The Eye)

Subset 7.1: Chromatic Map

- **Metric: Dominant Wavelength (nm)**
 - *Logic*: Mapping Low Frequencies to Long Wavelengths (Red) and High Frequencies to Short Wavelengths (Blue/Violet).
 - *Scoring*:
 - **Sub-Bass (40-60Hz)**: ~700nm (Deep Red)
 - **Kick/Low-Mid (60-250Hz)**: ~600nm (Orange/Amber)
 - **Mids (250Hz-2kHz)**: ~520nm (Green/Teal)
 - **Highs (2kHz+)**: ~450nm (Blue/Purple/UV)

Section 8: Kinetic Pillar (The Body)

Subset 8.1: Biometric Entrainment

- **Metric: Entrainment Factor (0-100)**
 - *Logic*: Beat Salience (clarity of the pulse) combined with Repetition.
 - *Scoring*:
 - < 30: Ambient/Drone (No motor response).
 - 30-70: Head Nod (Complex rhythms).
 - 70: Body Lock (Four-on-the-floor dance).
- **Metric: MET Score (Metabolic Equivalent)**
 - *Logic*: BPM mapped to physical exertion levels.
 - *Scoring*:
 - **1.0 MET**: 0-60 BPM (Rest)
 - **3.0 MET**: 80-100 BPM (Walk)
 - **8.0+ MET**: 140+ BPM (Sprint/HIIT)

Section 9: Genealogical Pillar (The Roots)

Subset 9.1: Tribe Alignment (GVS)

- **Metric: Authenticity Ratio (GVS)**
 - *Logic*: $(\text{Positive Votes by "Tribe Members"} / \text{Total Votes})$.
 - *Scoring*:
 - "Tribe Member" defined as user with >50 hours listening to that specific sub-genre.

- **True:** > 80% approval from Tribe.
- **Mainstream/Poser:** < 40% approval from Tribe.